

Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the **cSEM** package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A . Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load **cSEM**, specify the model using **lavaan**-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

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1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrandal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004).

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schuberth, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schuberth, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982), also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schuberth, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schuberth, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schuberth, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schuberth, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as *matrixpls* (Rönkkö, 2022), *SEM-inR* (Ray, Danks, & Calero Valdez, 2026), *cSEM* (Rademaker & Schubert, 2020) and *plspm* (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as *plspm* (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (UNESCO, 2021). For example, their codebase is not openly accessible, which impedes transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package *cSEM* as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

2 cSEM R Illustration

2.1 Getting started with cSEM

To use the *cSEM* package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, refer to the official guides for R (<https://cran.r-project.org>) and RStudio (<https://posit.co/download/rstudio-desktop>).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click **File** → **New File** → **R Script**. The typical workflow is to write code in the editor and run the selected line(s) with **Ctrl + Enter**. Any line beginning with **#** is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.  
1 + 1 # Code is executed; the result is printed in the Console.  
  
[1] 2
```

Before *cSEM* can be used, it must be installed. The *cSEM* package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: <https://github.com/FloSchubert/cSEM>. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)
```

With `cSEM` installed and loaded, the data can now be imported and the model specified, as described in the following sections.

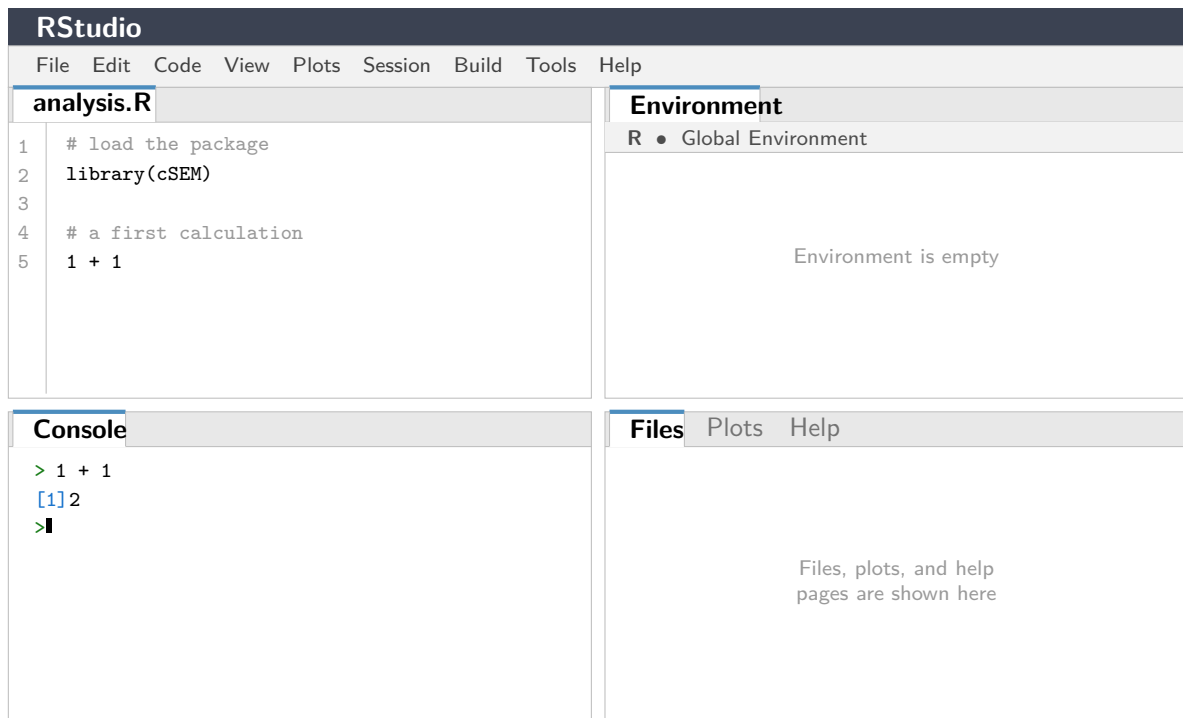


Figure 1: The four panes of the **RStudio** interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The `cSEM` package (Rademaker & Schuberth, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. Following Cheah et al. (2026), our illustration of the application of PLSc using `cSEM` draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13).

The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

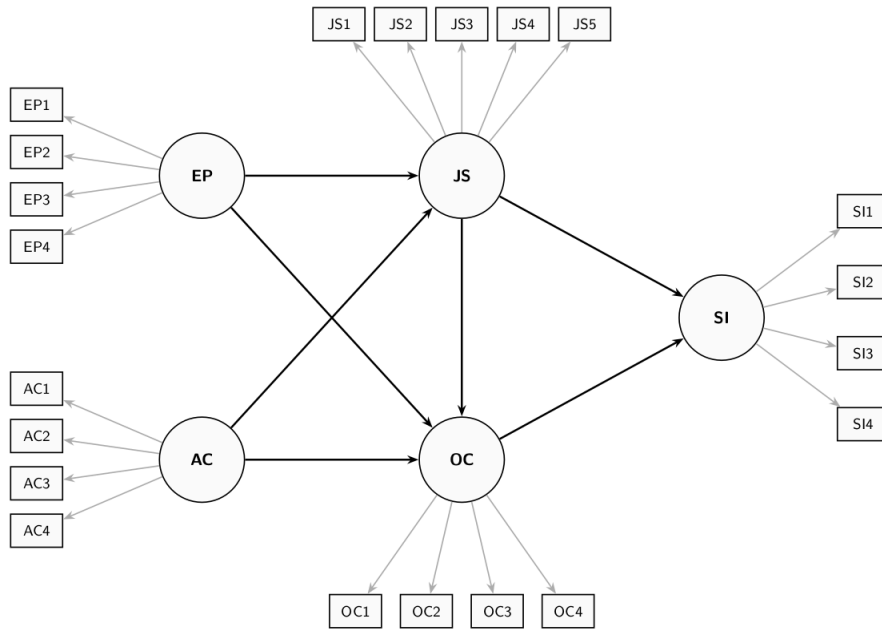


Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: <https://www.smartpls.com/documentation/sample-projects/employee-retention>. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, the data is assumed to be downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as a CSV file using ';' as the separator, thus it can be loaded into R using the following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                             "EP1", "EP2", "EP3", "EP4",
                             "JS1", "JS2", "JS3", "JS4", "JS5",
                             "OC1", "OC2", "OC3", "OC4",
                             "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400 21

# count missing values per variable
colSums(is.na(HBAT_SEM_rel))

AC1 AC2 AC3 AC4 EP1 EP2 EP3 EP4 JS1 JS2 JS3 JS4 JS5 OC1 OC2 OC3 OC4 SI1 SI2 SI3
0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
SI4
0
```

Since `cSEM` can only handle complete datasets, i.e., datasets without any missing values, the dataset is checked using the `is.na()` function. The output shows that the variables AC4

and EP4 each contain one missing value. To ensure that the estimation can be performed the missing observations need to be deleted or imputation methods need to be applied. Deleting incomplete observations can be done via the following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

# the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398  21
```

Since two observations contain missing values, all analyses in the remainder of this tutorial are based on the $N = 398$ complete observations. When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to **cSEM** functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In **cSEM**, models are specified using the **lavaan** model syntax. The **lavaan** package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The **cSEM** specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"
```

2.4 Model estimation

The central function is `csem()`. The minimal input required for the `csem()` function is a dataset and a model. The dataset should be a data frame or a matrix with column names matching the indicator names used in the model. The package **cSEM** handles a lot of different data types natively, such as "logical", "numeric" ("double" or "integer"), "factor" ("ordered" and/or "unordered"), "character" or a mix of several types. The model is needed to be written in **lavaan** syntax, with `=~` for measurement relations and `~` for structural (regression) relations.

By default weights are estimated using the partial least squares path modeling algorithm ("PLS-PM").

To ultimately estimate with the `csem` function, the function is called and save the output is saved as `res_csem`:

Inference and confidence intervals can be obtained via bootstrap resampling. For this the parameter `.resample_method` is set to "bootstrap", another option is "jackknife". Confidence intervals are calculated based on .R bootstrap resamples, and a `.seed` can be set to ensure reproducibility.

```
# PLSc-SEM estimation
res_csem <- csem(
  .data      = HBAT_SEM_rel,
  .model     = model,
  .resample_method = "bootstrap",
  .R         = 1000,
  .seed      = 42
)
```

Warning: package 'future' was built under R version 4.5.3

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```
# Full results summary
summary_res <- summarize(res_csem)
summary_res
```

```
----- Overview -----

General information:
-----
Estimation status      = Ok
Number of observations  = 398
Weight estimator       = PLS-PM
Inner weighting scheme = "path"
Type of indicator correlation = Pearson
...

----- Estimates -----

Estimated path coefficients:
=====
```

Path	Estimate	Std. error	t-stat.	p-value	CI_percentile 95%
JS ~ EP	0.2448	0.0518	4.7262	0.0000	[0.1449; 0.3475]
JS ~ AC	-0.0033	0.0598	-0.0549	0.9562	[-0.1164; 0.1189]
OC ~ EP	0.4288	0.0584	7.3384	0.0000	[0.3218; 0.5460]
...					

```
----- Effects -----
```



```
Estimated total effects:
```

```
=====
```

Total effect	Estimate	Std. error	t-stat.	p-value	CI_percentile 95%
JS ~ EP	0.2448	0.0518	4.7262	0.0000	[0.1449; 0.3475]
JS ~ AC	-0.0033	0.0598	-0.0549	0.9562	[-0.1164; 0.1189]
OC ~ EP	0.4523	0.0558	8.1108	0.0000	[0.3548; 0.5687]
...					

The output above has been shortened for readability. The "..." show omitted lines. Executing the provided code will return the whole output.

3 Assessment Methods

Next, the model estimates are assessed, according to the guidelines described in Benitez et al. (2020). This assessment comprises three parts: The assessment of the overall model fit, the assessment of the reflective measurement models, and the evaluation of the structural model.

3.1 Overall model fit

Before the overall model fit can be tested, the estimation needs to be validated. This is done with the `verify()` function. This function gives the user details about the estimation. It gives information if convergence is achieved, all absolute standardized loading estimates are smaller or equal than one. It also checks whether the construct variance covariance matrix is positive semi-definite, if all reliability estimates are smaller or equal than one and if the model-implied indicator variance covariance matrix is positive semi-definite. If a check fails, the results should not be interpreted and the user needs to customize the algorithm to ensure a valid estimation process. For that check Section 4.1.

```
# Check whether the estimation is admissible
```

```
verify(res_csem)
```

```
Verify admissibility:
```

```
admissible
```

```
Details:
```

Code	Status	Description
1	ok	Convergence achieved
2	ok	All absolute standardized loading estimates <= 1
3	ok	Construct VCV is positive semi-definite
4	ok	All reliability estimates <= 1
5	ok	Model-implied indicator VCV is positive semi-definite

After that the overall model fit with the function `testOMF()` can be examined.. The function `testOMF()` is a bootstrap-based test for model fit originally proposed by Beran and Srivastava

(1985). See also Dijkstra and Henseler (2015a) who first suggested the test in the context of PLS-PM. By default `testOMF()` tests the null hypothesis that the population indicator correlation matrix equals the population model-implied indicator correlation matrix. `testOMF()` uses different distance measures to assess the distance between the sample indicator correlation matrix and the estimated model-implied indicator correlation matrix, such as the geodesic distance d_G , the standardized root mean square residual $SRMR$, the squared Euclidean distance d_L and the distance based on the maximum likelihood fit function d_{ML} . The reference distribution for each test statistic is obtained via bootstrap with `.R` resamples as proposed by Beran and Srivastava (1985). Reproducibility is ensured through the parameter `.seed`.

```
# Bootstrap-based test of the overall model fit
```

```
fit_test <- testOMF(
  res_csem,
  .R          = 1000,
  .seed       = 42,
  .handle_inadmissibles = "replace"
)
```

```
fit_test
```

```
----- Test for overall model fit based on Beran & Srivastava (1985) -----
```

```
Null hypothesis:
```

```
+-----+
|                                     |
|  H0: The model-implied indicator covariance matrix equals the  |
|  population indicator covariance matrix.                        |
|                                     |
+-----+
```

```
Test statistic and critical value:
```

Distance measure	Test statistic	Critical value
		95%
dG	0.3253	0.3390
SRMR	0.0706	0.0506
dL	1.1505	0.5913
dML	1.8972	2.0221

```
Decision:
```

Distance measure	Significance level
	95%
dG	Do not reject
SRMR	reject
dL	reject
dML	Do not reject

```
Additional information:
```

```
Out of 2086 bootstrap replications 1000 are admissible.
```

```
See ?verify() for what constitutes an inadmissible result.
```

```
The seed used was: 42
```

3.2 Assessment of the reflective measurement models

The next step is about assessing the reflective measurement models. The assessment follows the guideline described in Benitez et al. (2020). It includes reliability, indicator reliability, convergent validity, and discriminant validity. The `assess()` function computes all relevant quality criteria. The output shows all criteria grouped in titled sections. The output here is shortened for presentation, the `"..."` mark omitted lines, running the code yields the full report.

```
#Compute all quality criteria
quality <- assess(res_csem)
quality
```

```
-----
Construct      AVE      R2      R2_adj
EP             0.6103      NA      NA
AC             0.6778      NA      NA
JS             0.5203      0.0595    0.0548
...

----- Common (internal consistency) reliability estimates -----
Construct Cronbachs_alpha  Joereskogs_rho  Dijkstra-Henselers_rho_A
EP          0.8596          0.8607          0.8717
AC          0.8936          0.8919          0.9105
JS          0.8450          0.8417          0.8568
...

----- Alternative (internal consistency) reliability estimates -----
Construct      RhoC      RhoC_mm      RhoC_weighted
EP             0.8607      0.8557      0.8717
AC             0.8919      0.8765      0.9105
JS             0.8417      0.8267      0.8568
...

----- Distance and fit measures -----
Geodesic distance      = 0.3252628
Squared Euclidean distance = 1.150488
ML distance            = 1.89721
...

----- Model selection criteria -----
Construct      AIC      AICc      AICu
```

```

JS          -19.4331      380.6687      -16.4217
OC          -125.2111      274.9420      -121.1909
SI          -128.2165      271.8852      -125.2052
...

----- Variance inflation factors (VIFs) -----

Dependent construct: 'JS'

Independent construct    VIF value
EP                      1.0695
...

----- Effect sizes (Cohen's f^2) -----

Dependent construct: 'JS'

Independent construct      f^2
EP                       0.0596
...

----- Discriminant validity assessment -----

Heterotrait-monotrait ratio of correlations matrix (HTMT matrix)
-----

Values in the lower triangular part are the absolute HTMT values.
...

----- Effects -----

Estimated total effects:
=====

Total effect    Estimate    Std. error    t-stat.    p-value    CI_percentile
                95%
...

```

In the following, the criteria of `assess()` are extracted one at a time.

Evaluating the reliability of the construct scores

In this step we evaluate whether the items reliably represent the underlying construct. Reliability asks how much of the items variance can be explained by the underlying construct. A construct with high reliability tends to produce similar items under consistent conditions. The `cSEM` package offers a variety of different reliability coefficients. These include Cronbach's α , Jöreskog's congeneric reliability ρ_C , and Dijkstra-Henseler's ρ_A . These coefficients differ in their assumption imposed on the measurement models. Cronbach's α assumes ... weights and ... items. Jöreskog's ρ_C allows loadings to ... but still assumes ... weights. Dijkstra-Henseler's ρ_A ...

We can extract the information for the reliability assessment out of the `assess` object with the `$` operator:

```
#Show Dijkstras and Henselers rho A
quality$Reliability

$Cronbachs_alpha
      EP      AC      JS      OC      SI
0.8595716 0.8936315 0.8450310 0.8328395 0.8890480

$Joereskogs_rho
      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

$`Dijkstra-Henselers_rho_A`
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

This prints out Cronbach_alpha, Joereskogs_rho, Dijkstra-Henseler_rho_A. Additionnaly the `assess()` function offers alternative reliability coefficients.

```
# Show rho_C
quality$RhoC

      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

quality$RhoC_mm

      EP      AC      JS      OC      SI
0.8556731 0.8765202 0.8266506 0.8029492 0.8860215

quality$RhoC_weighted

      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122

quality$RhoC_weighted_mm

      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

Evaluating indicator reliability

Indicator reliability ensures that ... It can be assessed via the factor loadings ... For that we extract out of the csem object the `loading_estimates` and its standard errors and its corresponding confidence intervals. This requires the bootstrap or jackknife for resampling. Thats why we did the bootstrap in the call of the `csem` function.

```
# Loading estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem)$Estimates$Loading_estimates

      Name Construct_type Estimate Std_err t_stat p_value
1 EP =~ EP1 Common factor 0.6341011 0.08825730 7.184687 6.736133e-13
2 EP =~ EP2 Common factor 0.8786819 0.06817644 12.888350 5.235172e-38
3 EP =~ EP3 Common factor 0.7678021 0.08865615 8.660449 4.699094e-18
```

4	EP =~ EP4	Common factor	0.8230691	0.06234019	13.202864	8.446117e-40
5	AC =~ AC1	Common factor	0.7691299	0.08208943	9.369415	7.293568e-21
6	AC =~ AC2	Common factor	0.6765259	0.09899325	6.834061	8.254389e-12
7	AC =~ AC3	Common factor	0.8215860	0.07771581	10.571671	4.032442e-26
8	AC =~ AC4	Common factor	0.9933858	0.04413543	22.507672	3.491295e-112
9	JS =~ JS1	Common factor	0.6248337	0.11846811	5.274278	1.332798e-07
10	JS =~ JS2	Common factor	0.6964587	0.10384121	6.706959	1.987220e-11
11	JS =~ JS3	Common factor	0.7182274	0.10374878	6.922755	4.429422e-12
12	JS =~ JS4	Common factor	0.6319045	0.12166320	5.193883	2.059521e-07
13	JS =~ JS5	Common factor	0.9004003	0.09311164	9.670115	4.039225e-22
14	OC =~ OC1	Common factor	0.4252105	0.06655371	6.388983	1.669924e-10
15	OC =~ OC2	Common factor	0.9574682	0.03086106	31.025128	2.470773e-211
16	OC =~ OC3	Common factor	0.6858765	0.05274373	13.003943	1.161946e-38
17	OC =~ OC4	Common factor	0.8565670	0.04700870	18.221458	3.487351e-74
18	SI =~ SI1	Common factor	0.8170099	0.04630598	17.643722	1.137185e-69
19	SI =~ SI2	Common factor	0.8705194	0.03913952	22.241445	1.364890e-109
20	SI =~ SI3	Common factor	0.6778349	0.05957767	11.377332	5.423357e-30
21	SI =~ SI4	Common factor	0.8959341	0.04423967	20.251826	3.423134e-91
	CI_percentile.95%L	CI_percentile.95%U				
1		0.4532620	0.7798607			
2		0.7214431	0.9805613			
3		0.5959517	0.9257479			
4		0.6938317	0.9348749			
5		0.6214292	0.9414872			
6		0.4665006	0.8574168			
7		0.6735746	0.9734165			
8		0.8318356	0.9978215			
9		0.3740575	0.8383272			
10		0.4843479	0.8895589			
11		0.4989930	0.8887790			
12		0.3733224	0.8466869			
13		0.6369939	0.9889399			
14		0.2851689	0.5408111			
15		0.8851819	0.9963862			
16		0.5873668	0.7951513			
17		0.7585917	0.9460804			
18		0.7268224	0.9095172			
19		0.7801786	0.9365719			
20		0.5648917	0.7977440			
21		0.8073558	0.9718306			

Evaluating convergent validity

Convergent validity is the extent to which the indicators belong to one latent variable actually measure the same construct.

Average variance extracted (AVE)

quality\$AVE

	EP	AC	JS	OC	SI
	0.6102822	0.6777668	0.5202693	0.5754208	0.6718668

Evaluating discriminant validity

Discriminant validity is the extent to which the latent variables are enough to distinct of each other so it makes sense to model them as distinct variables. The **cSEM** package uses the HTMT and its improved version HTMT2 to assess discriminant validity. Both are calculated with the default options within the **assess()** function. If the user wants to do inference the **.inference** parameter can be set to "bootstrap" or "asymptotic" then the associated confidence intervals are being calculated. While doing inference it is recommended to set **.absolute** to FALSE. If one is bigger than the upper limit then discriminant validity is seen as established.

```
# Heterotrait-monotrait ratio of correlations (HTMT)
```

```
HTMT <- assess(res_csem,  
  .quality_criterion = "htmt",  
  .inference = "asymptotic",  
  .absolute = FALSE)
```

Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either
all-positive or all-negative.

HTMT

```
-----  
----- Discriminant validity assessment -----  
  
Heterotrait-monotrait ratio of correlations matrix (HTMT matrix)  
-----  
  
Values in the lower triangular part are the HTMT values.  
  
Values in the upper triangular part are the 95% limits of the  
asymptotic confidence intervals based on the delta-method.  
  
      EP      AC      JS      OC      SI  
EP 1.0000000 0.34892877 0.3266616 0.5811953 0.6575878  
AC 0.2555939 1.00000000 0.1432197 0.3575722 0.3955342  
JS 0.2437979 0.05239146 1.0000000 0.3056150 0.3131749  
OC 0.4939933 0.27449892 0.2086788 1.0000000 0.5886913  
SI 0.5672796 0.30936763 0.2313363 0.5005239 1.0000000  
-----
```

The warning message says that some of the correlations between the indicators have different signs. This does not lead to any calculation issues.

The improved version HTMT2 and its inference can be calculated via bootstrap confidence intervals.

```
# Heterotrai-monotrait ratio 2 of correlations (HTMT2)
```

```
HTMT2 <- assess(res_csem,  
  .quality_criterion = "htmt2",
```

```
.inference = "bootstrap",
.absolute = FALSE,
.seed = 42)
```

HTMT2

```
-----
----- Discriminant validity assessment -----
-----
Advanced heterotrait-monotrait ratio of correlations matrix (HTMT2 matrix)
-----

Values in the lower triangular part are the HTMT2 values.

Values in the upper triangular part are the 95%-quantiles of the
bootstrap confidence intervals (using .ci = 'CI_percentile')
based on 106 valid bootstrap runs.

          EP          AC          JS          OC          SI
EP 1.0000000 0.3668367 0.3164519 0.5915619 0.6592745
AC 0.2529206 1.0000000 0.1217016 0.3467054 0.3969569
JS 0.2376530      NaN 1.0000000 0.3014591 0.2951276
OC 0.4712082 0.2514075 0.1977162 1.0000000 0.5765823
SI 0.5661731 0.3083707 0.2219788 0.4705377 1.0000000
-----
```

Assessing multicollinearity among the indicators and Evaluating the weights

```
# VIF values for Mode B (composite) weights
quality$VIF_modeB
```

```
# Weight estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem)$Estimates$Weight_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP <~ EP1	Common factor	0.2425256	0.03033059	7.996073	1.284500e-15
2	EP <~ EP2	Common factor	0.3360708	0.03417091	9.834998	7.956905e-23
3	EP <~ EP3	Common factor	0.2936625	0.02620371	11.206907	3.771439e-29
4	EP <~ EP4	Common factor	0.3148005	0.02512880	12.527478	5.281529e-36
5	AC <~ AC1	Common factor	0.2707114	0.02974091	9.102324	8.841910e-20
6	AC <~ AC2	Common factor	0.2381175	0.03264047	7.295162	2.982994e-13
7	AC <~ AC3	Common factor	0.2891744	0.02794699	10.347249	4.306835e-25
8	AC <~ AC4	Common factor	0.3496430	0.01889744	18.502136	1.984503e-76
9	JS <~ JS1	Common factor	0.2223363	0.04059423	5.477043	4.324929e-08
10	JS <~ JS2	Common factor	0.2478229	0.03710726	6.678555	2.413101e-11
11	JS <~ JS3	Common factor	0.2555689	0.03888206	6.572926	4.933583e-11
12	JS <~ JS4	Common factor	0.2248524	0.04381154	5.132264	2.862779e-07
13	JS <~ JS5	Common factor	0.3203920	0.03598888	8.902527	5.458932e-19
14	OC <~ OC1	Common factor	0.1740754	0.02457108	7.084563	1.394839e-12


```

15 OC <~ OC2 Common factor 0.3919744 0.01869390 20.968041 1.284457e-97
16 OC <~ OC3 Common factor 0.2807885 0.02061076 13.623392 2.907433e-42
17 OC <~ OC4 Common factor 0.3506668 0.01644880 21.318690 7.616096e-101
18 SI <~ SI1 Common factor 0.2882324 0.01534997 18.777390 1.156254e-78
19 SI <~ SI2 Common factor 0.3071099 0.01550302 19.809683 2.456267e-87
20 SI <~ SI3 Common factor 0.2391329 0.01876727 12.742019 3.453427e-37
21 SI <~ SI4 Common factor 0.3160760 0.01634721 19.335168 2.717827e-83
  CI_percentile.95%L CI_percentile.95%U
1      0.1828087      0.2999783
2      0.2725098      0.4008700
3      0.2414227      0.3459511
4      0.2697347      0.3683087
5      0.2183269      0.3399151
6      0.1723162      0.2988541
7      0.2386440      0.3504807
8      0.2919151      0.3651913
9      0.1423157      0.3013401
10     0.1794724      0.3181615
11     0.1792914      0.3324710
12     0.1395502      0.3054149
13     0.2341224      0.3685690
14     0.1229501      0.2179109
15     0.3530282      0.4250516
16     0.2467718      0.3260126
17     0.3185797      0.3842689
18     0.2590976      0.3185720
19     0.2763323      0.3387036
20     0.2055149      0.2781864
21     0.2820537      0.3429069

```

3.3 Structural model evaluation

```

# Path coefficient estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem)$Estimates$Path_estimates

```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	JS ~ EP	Common factor	0.244824960	0.05180199	4.72616873	2.287956e-06
2	JS ~ AC	Common factor	-0.003282073	0.05980429	-0.05488023	9.562339e-01
3	OC ~ EP	Common factor	0.428781485	0.05842984	7.33839869	2.161644e-13
4	OC ~ AC	Common factor	0.172554413	0.04807177	3.58951689	3.312914e-04
5	OC ~ JS	Common factor	0.096069429	0.05433001	1.76825712	7.701793e-02
6	SI ~ JS	Common factor	0.131226588	0.04557423	2.87940336	3.984284e-03
7	SI ~ OC	Common factor	0.490013776	0.05420168	9.04056490	1.558642e-19
			CI_percentile.95%L	CI_percentile.95%U		
1			0.144931207	0.3475086		
2			-0.116359796	0.1188850		
3			0.321754615	0.5459673		
4			0.093157078	0.2698075		
5			-0.006019847	0.2047949		
6			0.046575779	0.2207631		
7			0.392550356	0.6053730		

```
# Effect sizes ( $f^2$ )
```

```
quality$F2
```

	EP	AC	JS	OC	SI
JS	0.05959141	1.070949e-05	0.00000000	0.00000000	0
OC	0.22615825	3.880845e-02	0.01209977	0.00000000	0
SI	0.00000000	0.000000e+00	0.02299583	0.3206432	0

```
# Coefficient of determination ( $R^2$ ) and adjusted  $R^2$ 
```

```
quality$R2
```

	JS	OC	SI
	0.05954032	0.28264578	0.28445571

```
quality$R2_adj
```

	JS	OC	SI
	0.05477849	0.27718370	0.28083270

4 Further features of cSEM

4.1 Customization of PLS-PM

4.2 predictive model assessment

4.3 non linear

5 Concluding remarks

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